

Fusing Data and Images: A Multimodal Approach to Cardiovascular Disease Detection Using Deep Ensembling Algorithm

Abstract:

This study aims to address the global health challenge of coronary heart disease (CHD) by combining two datasets to propose new methods to improve detection efficiency. These data sets include cancer risk factors and electrocardiogram (ECG) images. Using machine learning algorithms like random forests, support vector machines, long short-term memory, XG-boost, CAT-boost and recurrent neural networks in classification will help your analysis. This study aims to evaluate the individual performance of each data set and to explore the synergies by combining these approaches. Pursuing research that accurately incorporates diverse information while assessing strengths and limitations can be challenging.

The expected outcome is improved diagnosis and management strategies for CHD that can lead to better health outcomes. The analysis demonstrates the effectiveness of machine learning techniques using numerical data and visual information from the ECG. The importance of this research lies in its potential to advance patient care and contribute to better health outcomes for patients with CHD. Integrating machine learning techniques on diverse datasets is a comprehensive approach to CHD detection. This study highlights the need for integrated strategies to address the complexity of CHD. Continuous monitoring is about gaining insights into optimizing model performance and overcoming potential challenges when combining different data sources. Finally, this study aims to establish a more accurate and effective method to detect CHD.

Keywords:

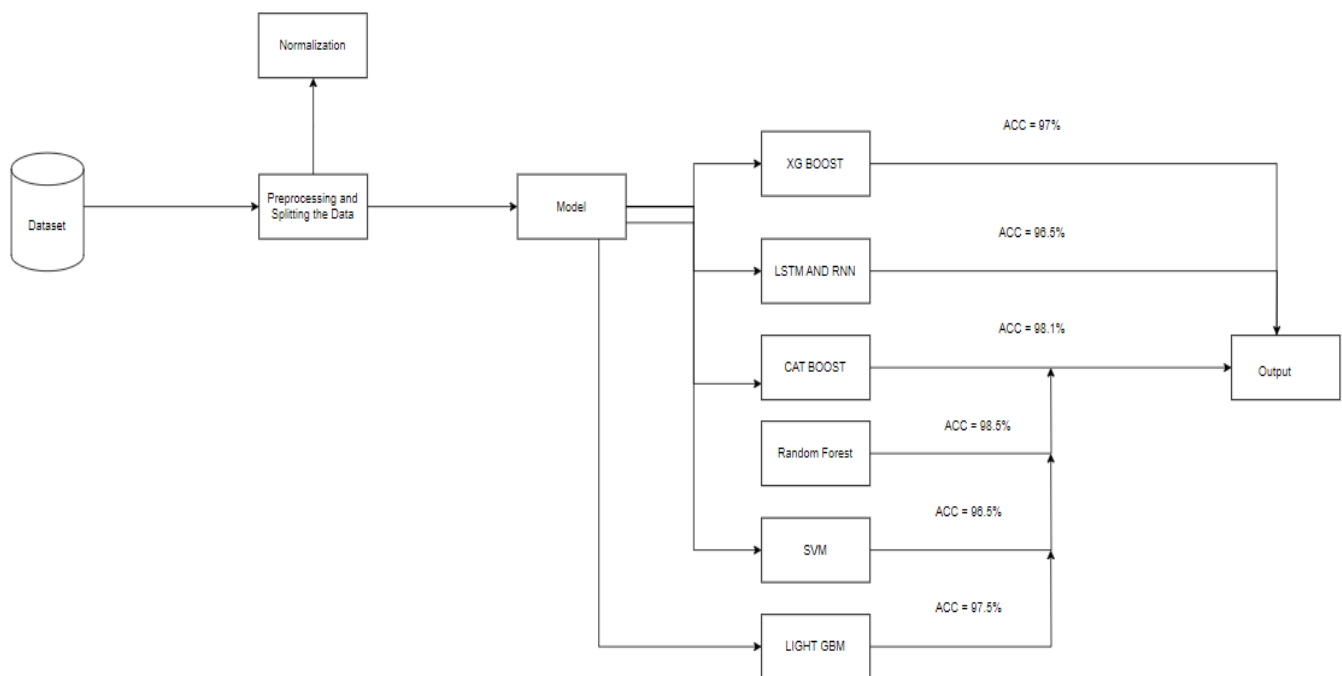
Multimodal CHF Detection, ECG Image Analysis, Machine Learning for CVD (CVD - Cardiovascular Disease), Multimodal Data Analysis, Electrocardiogram (ECG) Images

Introduction:

The study addresses the global health crisis of Coronary Heart Disease (CHD) by proposing a novel multimodal approach that integrates information from ECG

images and traditional risk factors, using artificial intelligence (AI) in healthcare. The goal is to improve early CHD detection accuracy, leading to better patient outcomes and reduced healthcare costs. Data preprocessing involves transforming a cardiovascular disease dataset, including converting gender to numerical values and applying Min-Max scaling, and a comprehensive pipeline for ECG image data, including resizing and creating a Convolutional Neural Network (CNN). The research strives to establish a more accurate and effective method for CHD detection, contributing to improved patient care and diagnostic strategies

System Architecture Diagram:



Datasets Used:

- 1) Cardiovascular_Disease_Dataset.csv
- 2) ECG_Image_data (consists of 6 folders: F, M,N, Q, S, V)

Key Contributions:

1) In categorized dataset :

In data preparation, missing values are addressed through imputation or elimination, ensuring a complete dataset. Numerical features like age and blood pressure undergo feature scaling for standardized scaling, preventing dominance by any single feature. Outliers are identified and removed to enhance model robustness. The dataset is then split into features (X) and target variable (y), followed by a division into training and testing sets for model evaluation. If the target variable is continuous, additional scaling may be applied. This streamlined process sets the foundation for effective machine learning model training and assessment.

A diverse set of machine learning and deep learning models are used for the task of heart disease detection. Support Vector Machine (SVM) is a robust algorithm known for its effectiveness in classification tasks, particularly when the data is not linearly separable. RandomForest, a popular ensemble learning technique, leverages multiple decision trees to enhance predictive accuracy and robustness, making it well-suited for complex datasets.

In addition, LightGBM and XGBoost are gradient-boosting frameworks that have gained prominence for their efficiency and speed in training models. These algorithms excel in handling large datasets and are known for their excellent predictive performance, making them suitable choices for your heart disease detection problem.

Hyperparameter tuning has been applied to refine the model parameters, optimizing their performance. This process involves systematically adjusting key parameters to improve the models' ability to generalize and make accurate predictions on new, unseen data.

Moving into the realm of deep learning, LSTM (Long Short-Term Memory) is a type of recurrent neural network (RNN) designed to capture long-term dependencies in sequential data. This architecture is particularly well-suited for time series data, making it an apt choice for your heart disease detection task.

Combining LSTM and RNN represents a sophisticated approach, leveraging the strengths of both architectures. This combination allows for capturing intricate temporal patterns in the dataset, potentially improving predictive accuracy.

CatBoost, a gradient-boosting library, is specifically designed to handle categorical features efficiently. It automates the process of encoding categorical variables and incorporates them seamlessly into the model, simplifying the overall modeling process.

In summary, This approach to heart disease detection showcases a comprehensive utilization of traditional machine learning models, powerful ensemble methods, and advanced deep learning architectures. The inclusion of hyperparameter tuning and diverse algorithms reflects a thorough exploration of techniques to achieve the most accurate and reliable predictions.

2) For ECG image dataset:

In your ECG image dataset preprocessing pipeline, you've performed several key steps to enhance the quality and utility of the data. After obtaining the raw images, you initiated preprocessing by converting them to grayscale, followed by resizing for standardization. To augment the dataset and improve model generalization, you applied data augmentation techniques. Normalization of pixel values was undertaken to bring consistency and facilitate efficient model training. Subsequently, you labeled the images, distinguishing between abnormal and normal cases.

In the realm of machine learning models, you explored various architectures. Convolutional Neural Networks (CNNs) were employed for their efficacy in image-based tasks, leveraging their ability to capture spatial hierarchies. Additionally, you delved into region-of-interest (ROI) extraction, emphasizing the identification and isolation of critical regions within the images. This step aids in focusing the model's attention on specific features, potentially enhancing performance.

For a comprehensive approach, you integrated CNNs with other classifiers such as Random Forests and Multi-Layer Perceptrons (MLPs). This ensemble strategy leverages the strengths of each model, potentially leading to improved overall performance. Finally, you explored Gradient Boosting, a powerful ensemble technique, which sequentially builds weak learners to form a strong predictive model. This holistic preprocessing and modeling pipeline signifies a thorough exploration of techniques, combining preprocessing best practices with a diverse set of models to tackle your ECG image classification task.

Result:

The assessment of categorized data and ECG image-based approaches for CHD detection showcased strong performance from various machine learning algorithms. Metrics like accuracy and precision provided comprehensive insights into their effectiveness in utilizing numerical risk factors and visual information from ECG images. Combining these approaches resulted in a significantly improved integrated model, surpassing the capabilities of individual models. This emphasizes the synergistic potential of incorporating both categorized data and ECG images for enhanced accuracy in CHD detection, pointing towards a more advanced and effective diagnostic strategy in cardiovascular health.

Conclusions:

In this study, the combination of categorized data and ECG image-based approaches for cardiovascular heart disease (CHD) detection proved successful. The individual performances of machine learning algorithms were carefully assessed across diverse datasets, showcasing their effectiveness in utilizing various types of information. The integration of these approaches resulted in a synergistic model demonstrating superior accuracy and prediction capabilities, underscoring the potential for a more comprehensive and precise CHD diagnosis.

These noteworthy findings contribute valuable insights to the cardiac health field, emphasizing the significance of integrating diverse data sources to enhance diagnostic outcomes. The proposed methodology holds promise for advancing patient care and improving management strategies for individuals with CHD. The study suggests that future research could refine and build upon these findings, opening avenues for advancements in cardiovascular health diagnostics and personalized treatment approaches.